

NETWORKING PROJECT

SETUP AND CONFIGURATION OF A THREE-LAN NETWORK USING A SINGLE ROUTER

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Project Deliverables

Objectives:

To configure and organize a structured three-LAN network using a single router, enabling efficient communication between separate network segments while applying proper IP addressing and basic routing principles. This project aims to strengthen practical understanding of network segmentation, device configuration, and real-world network design.

Key Component used in this Project:

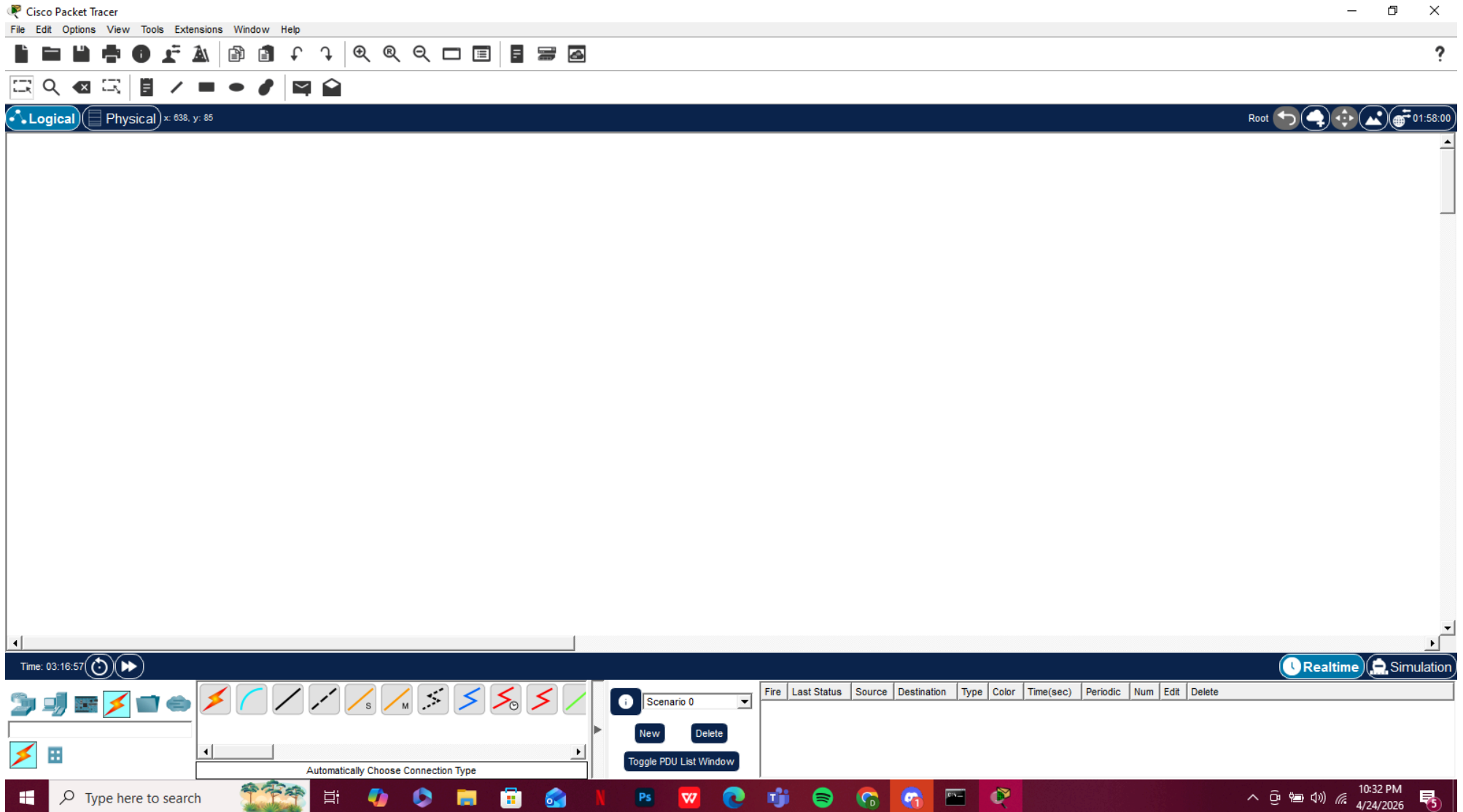
Component:	Description
● Cisco Packet Tracer ● Router ● DHCP Server ● Switch ● PCs & Laptop ● IP Addressing ● Cable	A network simulation tool used to design, configure, and test network setups virtually. Connects multiple networks together and directs data between them. Automatically assigns IP addresses and network settings to devices on a network. Connects devices within the same LAN and forwards data to the correct device. End-user devices used to send, receive, and test network communication. A system used to assign unique addresses to devices for identification on a network. Physical medium (e.g., Ethernet cable) used to connect network devices together.

Network Design:

LAN	Network Address	Devices
LAN 1	192.168.1.10	DHCP Server, 1 PCs. 2 laptop, 1 Switch
LAN 2	10.0.0.25	DHCP Server, 2 laptop, 1 Switch
LAN 3	172.32.5.50	DHCP Server, 1 PCs. 2 laptop, 1 Switch

Step 1:

Accessing the Cisco Packet Tracer Workspace after a successful login.

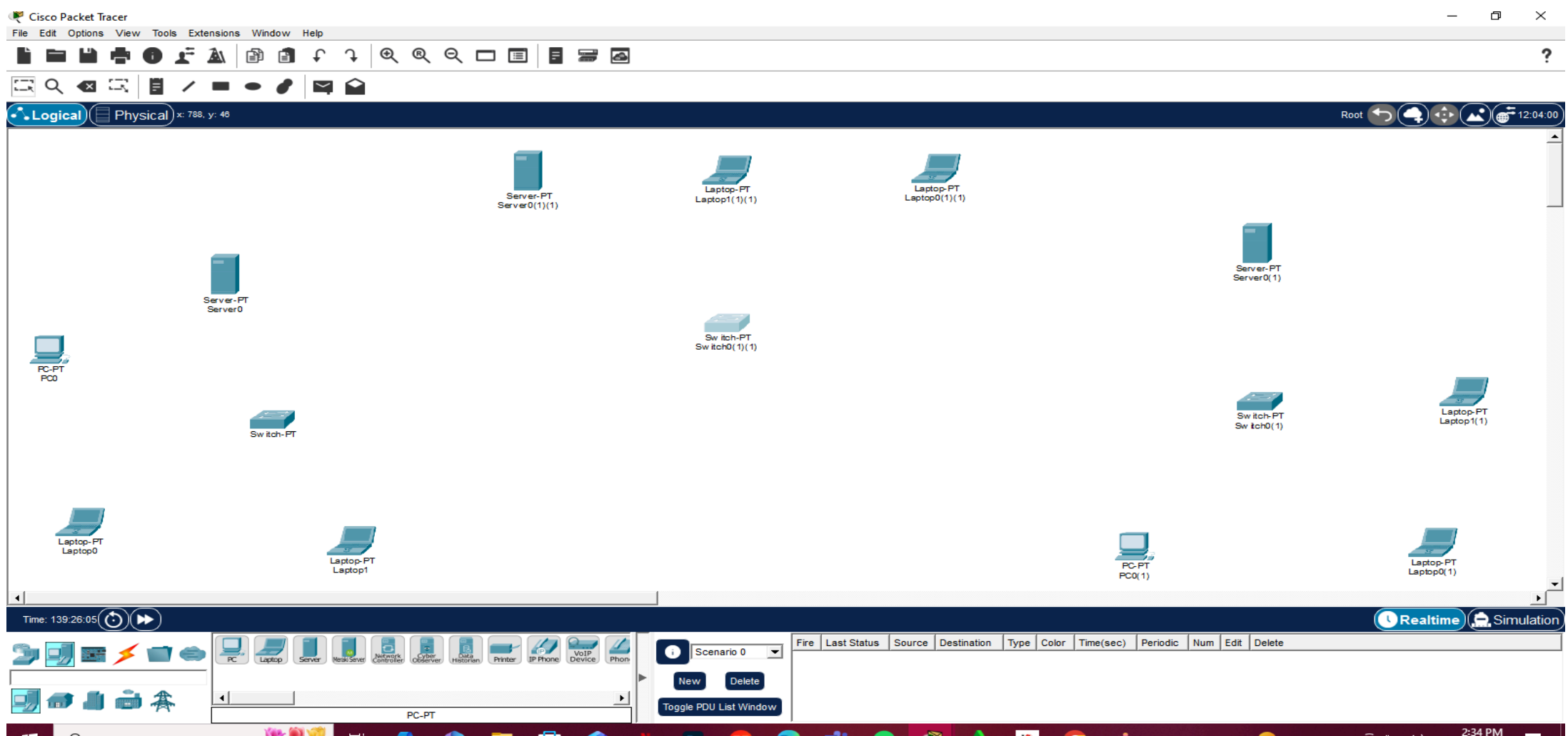


Step 2:

Add Network to Workspace:

In this step, network devices (PCs, Laptop, DHCP Server, Switch, Router) were added to the workspace using the device-type selection panel in the simulation tool.

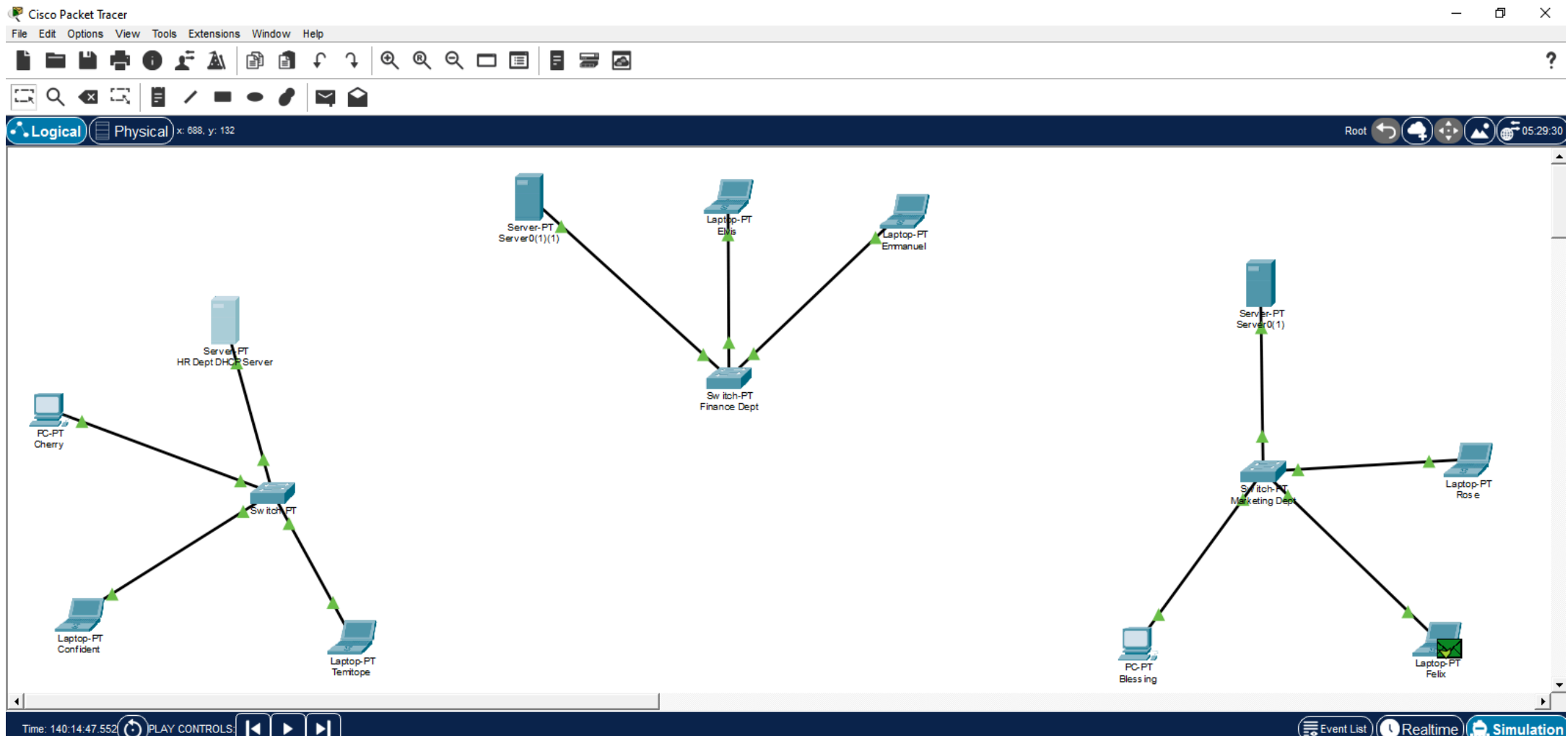
These devices were then dragged and dropped into the workspace. Three separate LANs were created by grouping the devices with their respective switches.



Step 3

Connect Network Devices:

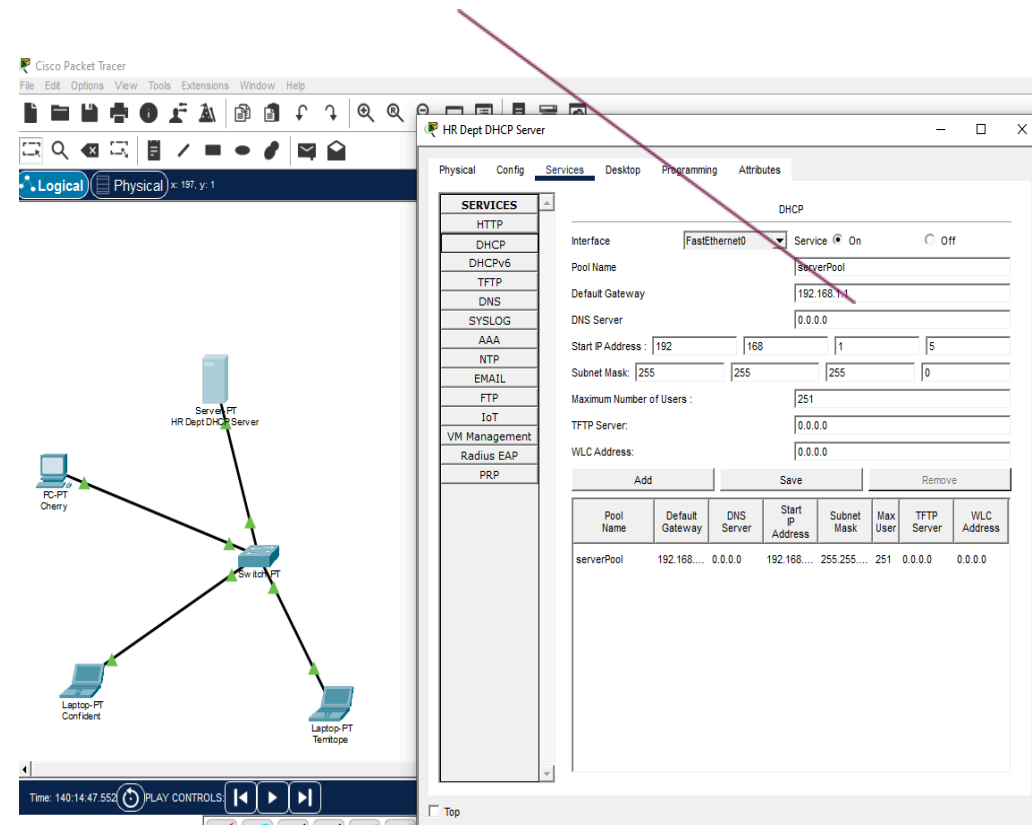
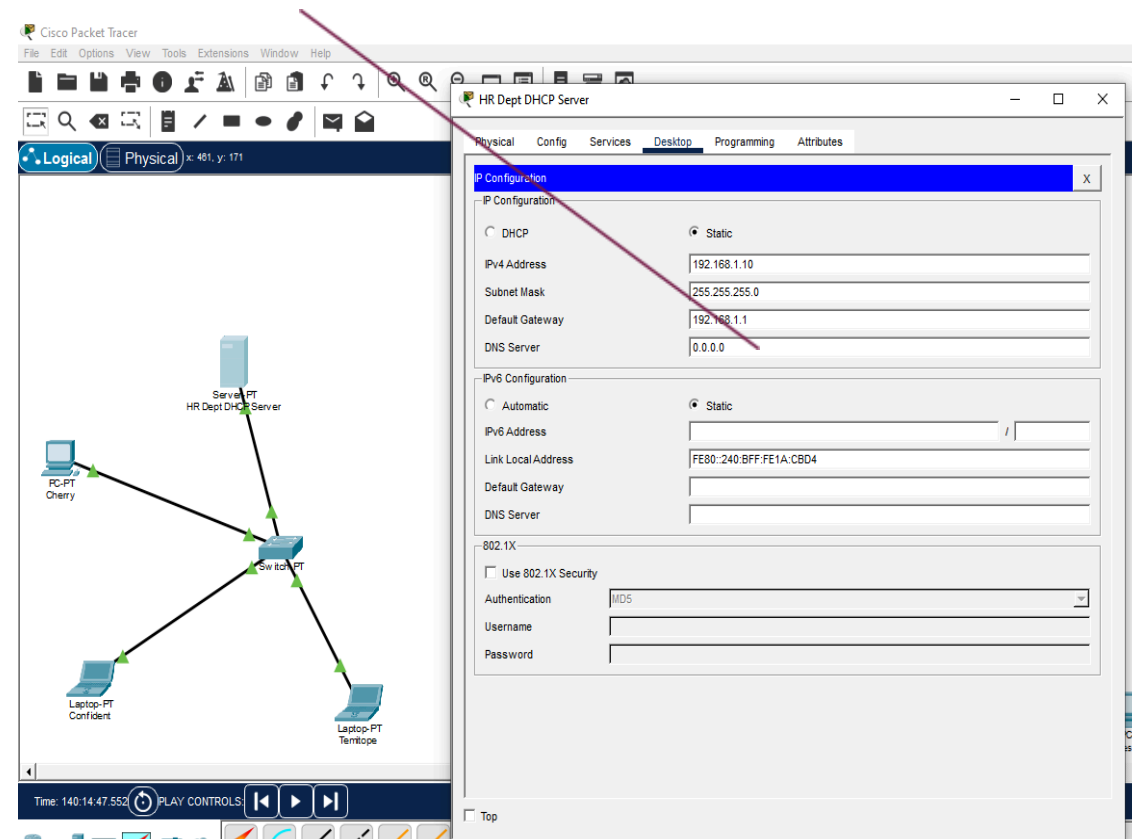
A Copper Straight-Through cable was used to connect the devices to their respective switches. Each LAN forms a star topology, where multiple devices connect to a central switch. The connection was created by selecting a cable tool from the bottom panel.



Step 4- LAN 1 DHCP SERVER CONFIGURATION

Assign Static IP and Configure DHCP Server:

A static IP address was assigned to the server and the DHCP server was configured to automatically assign IP addresses to other devices in the network. The server was configured manually via > desktop>IP configuration> static, entering an IP address -192.168.1.10, Subnet mask -255.255.255.0 and Default gateway -192.168.1.1. This ensures the server always has a consistent address in the network. After that DHCP service was enabled on the server by navigating to the DHCP configuration tab. A DHCP pool was created by specifying the IP address, Subnet mask, default gateway, starting IP address range for clients.



Step 5 -LAN 1

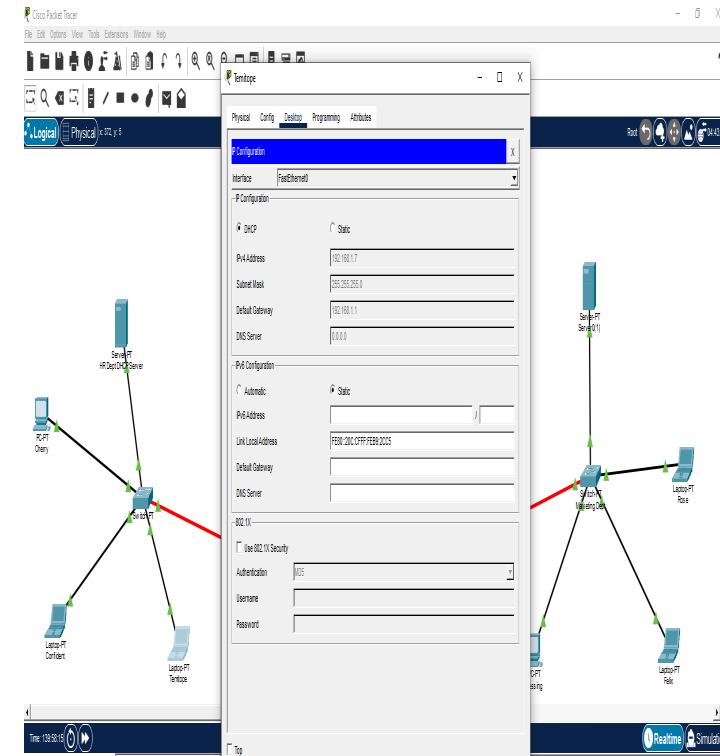
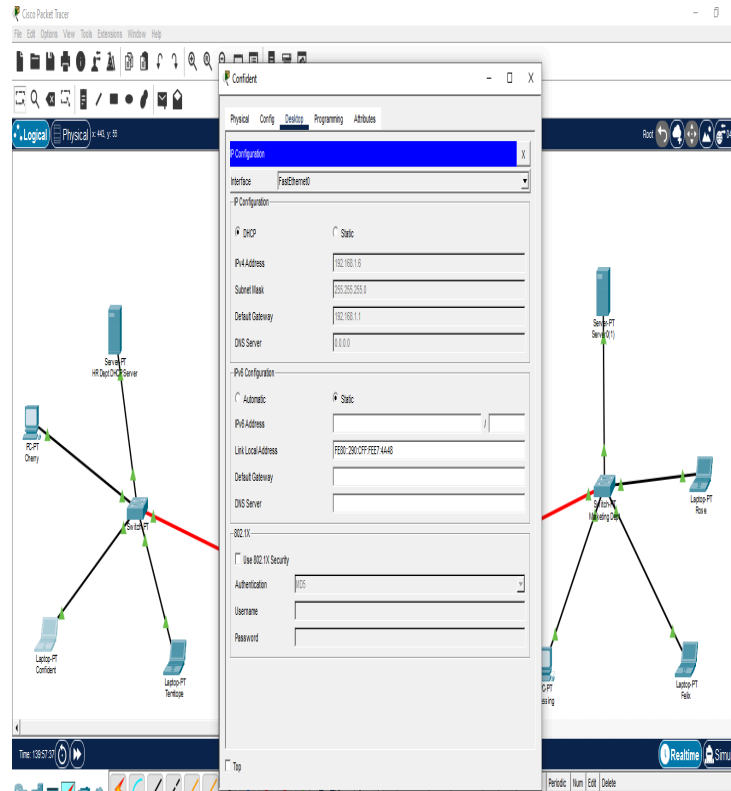
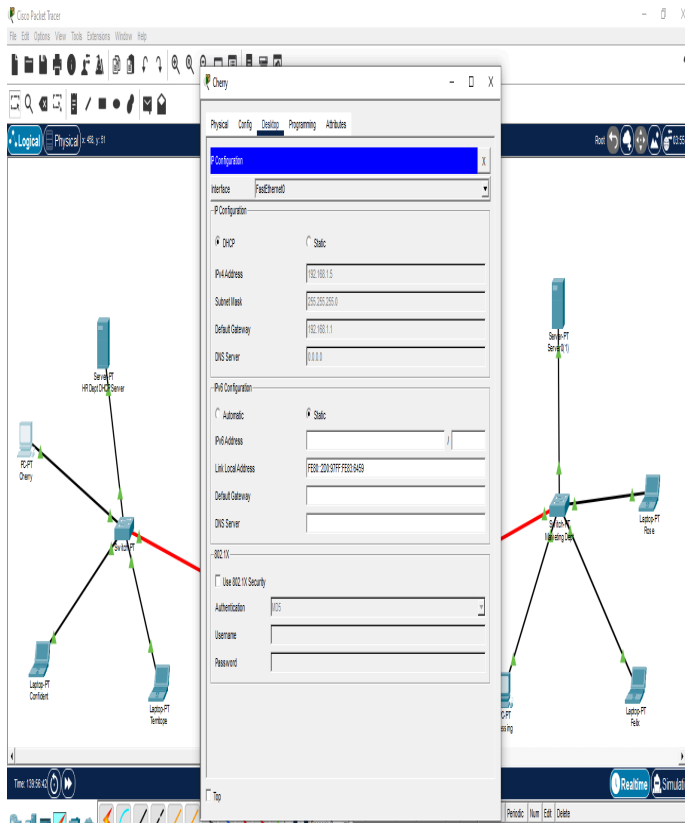
Verify DHCP IP Assignment:

All devices in LAN 1 were automatically assigned IP address, subnet mask, default gateway after the DHCP Configuration. Class C IP Address.

The first device(PC)
Successfully obtain IP address
Of 192.168.1,5 indicating proper
Communication with the DHCP
Server

The second device(laptop)
successfully obtain IP address
of 192.168.1.6 indicating proper
communication with the DHCP
server

The third device (Laptop)
obtained IP address of
192.168.1.7 indicating proper
communication with the
DHCP server

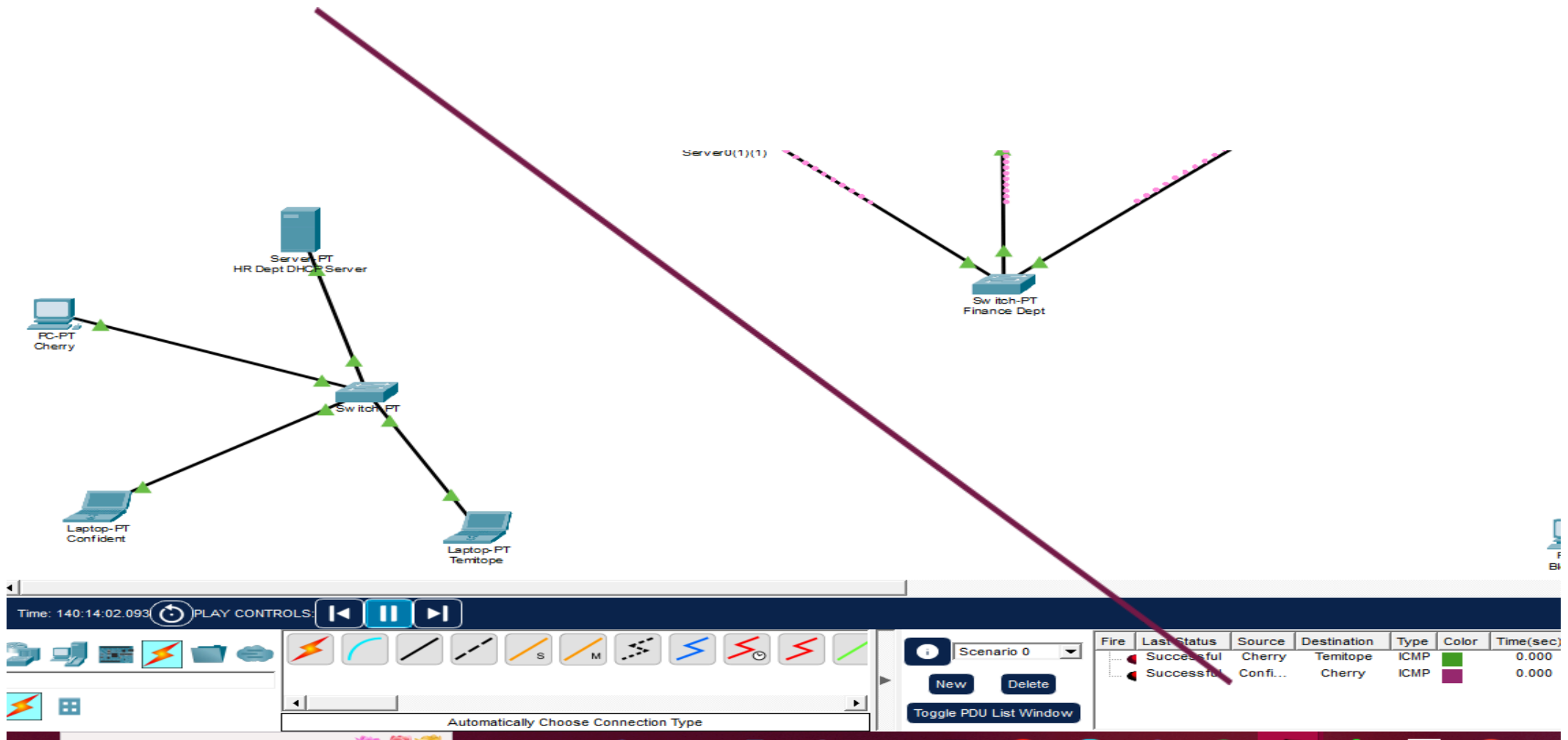


Step 6

Connectivity Test within LAN 1

To verify network communication within LAN 1, a connectivity test was performed by sending ping requests between devices.

Each device successfully responded to the ping, confirming that all hosts within the LAN 1 can communicate with each other.

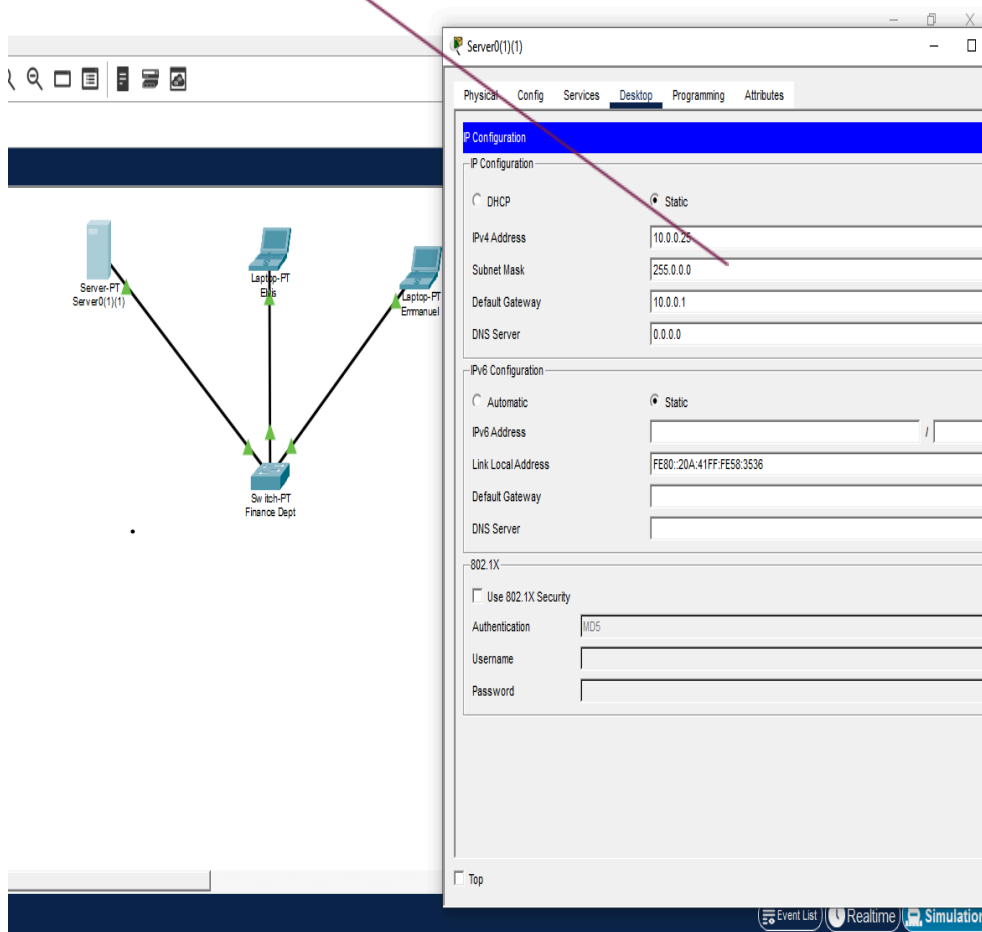


Step 7- LAN 2 DHCP SERVER CONFIGURATION

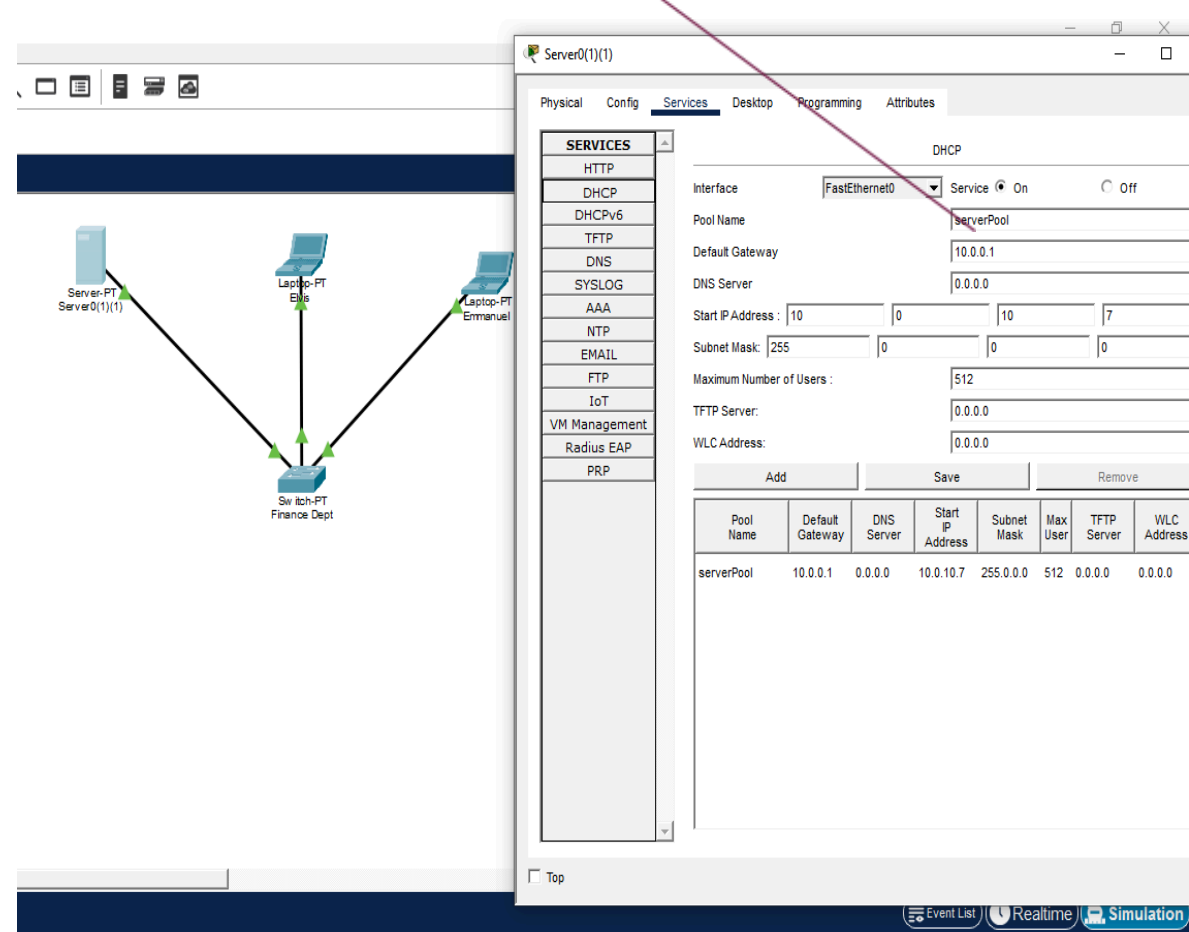
Assign Static IP and Configure DHCP Server:

A static IP address was assigned to the server and the DHCP server was configured to automatically assign IP addresses to other devices in the network. The server was configured manually via > desktop>IP configuration> static, entering an IP address -10.0.0.25 Subnet mask -255.0.0.0 and Default gateway -10.0.0.1. This ensures the server always has a consistent address in the network

After that DHCP service was enabled on the server by navigating to the DHCP configuration tab. A DHCP pool was created by specifying the IP address, Subnet mask, default gateway, starting IP address range for clients



The left panel shows a network diagram with a central switch labeled 'Sw Itch-PT Finance Dept' connected to three devices: 'Server-PT Server0(1)(1)', 'Laptop-PT Ellis', and 'Laptop-PT Emmanuel'. The right panel shows the 'Server0(1)(1)' configuration window with the 'Desktop' tab selected. The 'IP Configuration' section is set to 'Static' with the following values: IPv4 Address: 10.0.0.25, Subnet Mask: 255.0.0.0, Default Gateway: 10.0.0.1, and DNS Server: 0.0.0.0. The 'IPv6 Configuration' section is also set to 'Static' with a Link Local Address of FE80::20A:41FF:FE58:3536. The '802.1X' section is currently disabled.



The left panel shows the same network diagram as the previous image. The right panel shows the 'Server0(1)(1)' configuration window with the 'Services' tab selected. The 'DHCP' service is enabled. The 'Interface' is set to 'FastEthernet0'. The 'Pool Name' is 'ServerPool'. The 'Default Gateway' is 10.0.0.1 and the 'DNS Server' is 0.0.0.0. The 'Start IP Address' is 10.0.0.10 and the 'Subnet Mask' is 255.0.0.0. The 'Maximum Number of Users' is 512. The 'TFTP Server' and 'WLC Address' are both 0.0.0.0. Below the configuration fields is a table of DHCP pools:

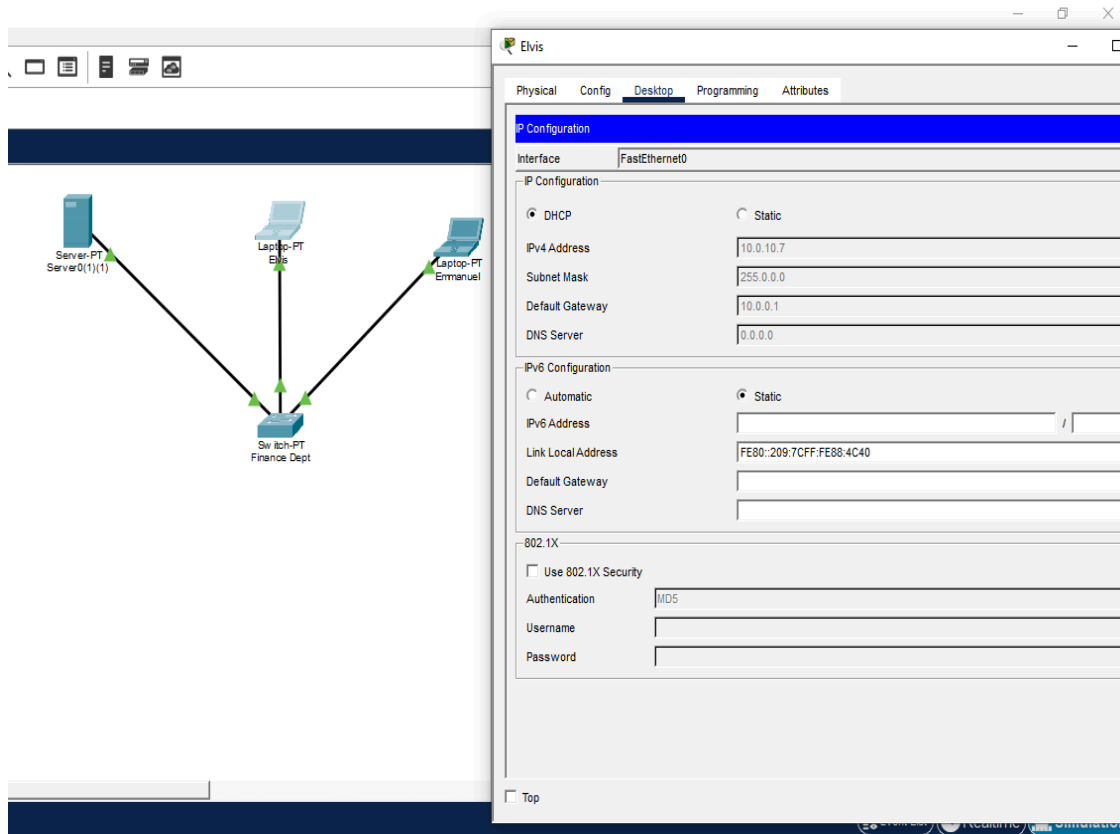
Pool Name	Default Gateway	DNS Server	Start IP Address	Subnet Mask	Max User	TFTP Server	WLC Address
serverPool	10.0.0.1	0.0.0.0	10.0.0.10	255.0.0.0	512	0.0.0.0	0.0.0.0

Step 8 -LAN 2

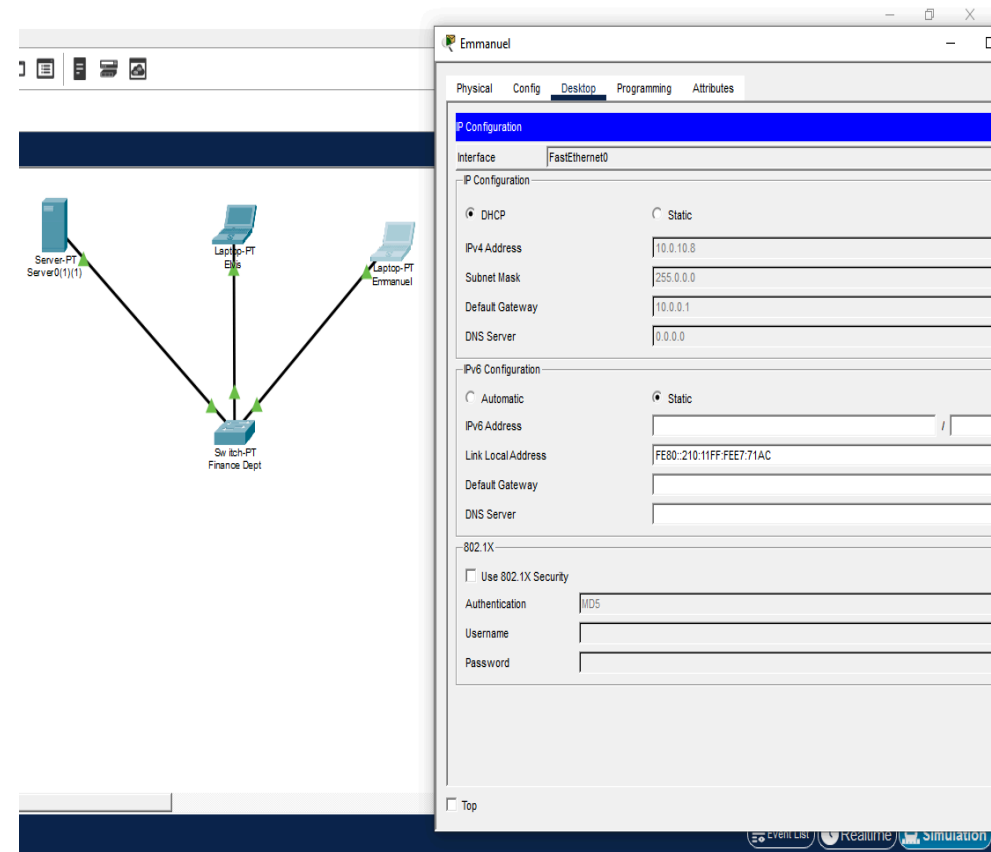
Verify DHCP IP Assignment:

All devices in LAN 2 were automatically assigned IP address, subnet mask, default gateway after the DHCP Configuration. Class A IP Address.

The first device(Laptop)
Successfully obtain IP address
Of 10.0.10.7 indicating proper
Communication with the DHCP
Server



The second device(laptop)
successfully obtain IP address
of 10.0.10.8 indicating proper
communication with the DHCP
server

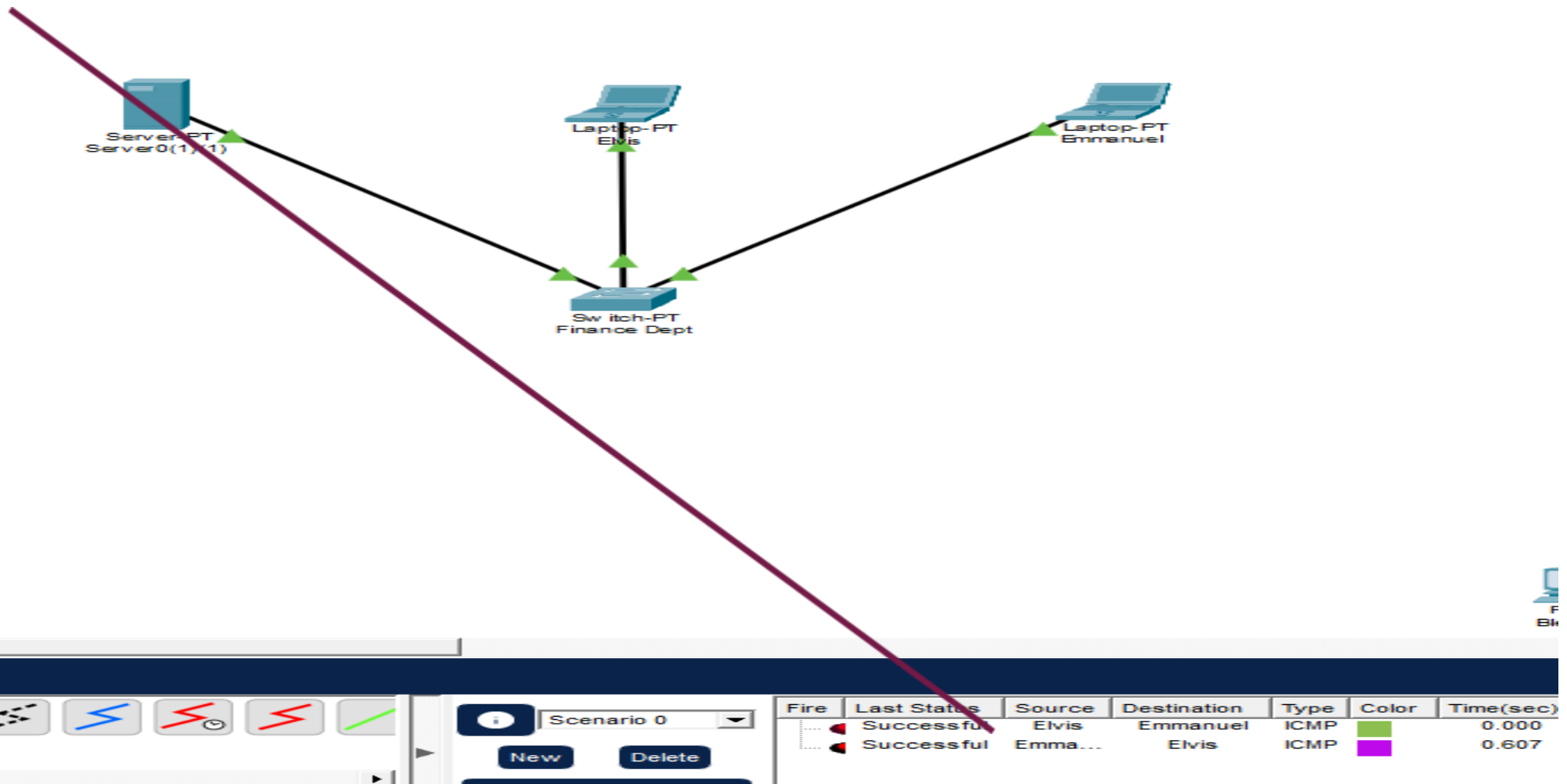


Step 9 LAN 2

Connectivity Test within LAN 2

To verify network communication within LAN 2, a connectivity test was performed by sending ping requests between devices.

Each device successfully responded to the ping, confirming that all hosts within the LAN 2 can communicate with each other.

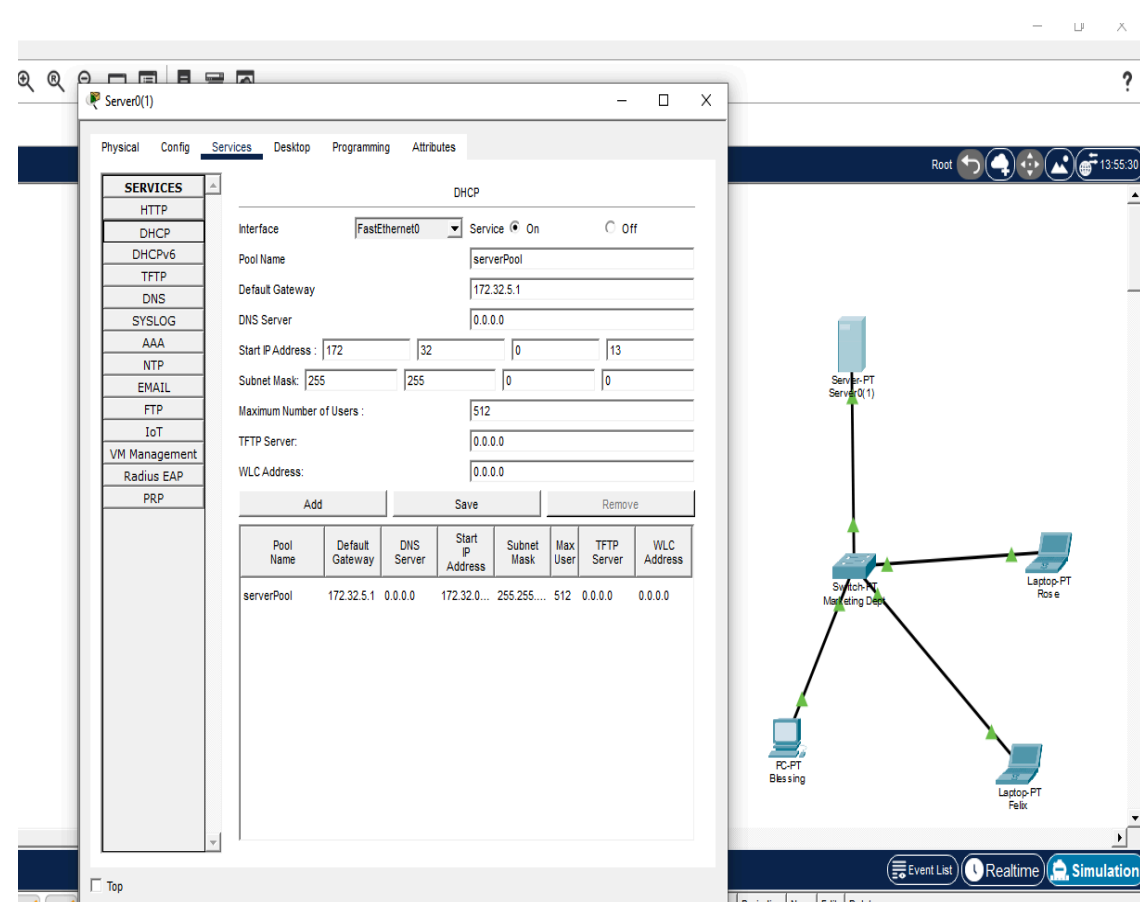
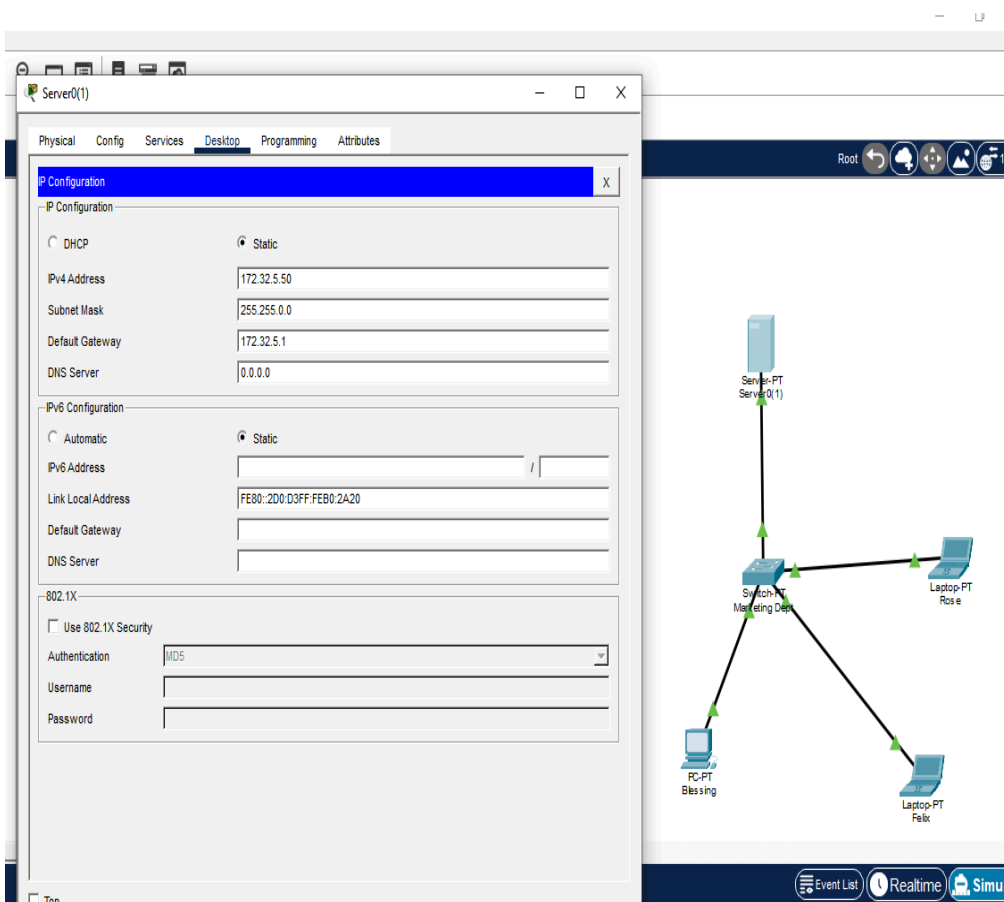


Step 10- LAN 3 DHCP SERVER CONFIGURATION

Assign Static IP and Configure DHCP Server:

A static IP address was assigned to the server and the DHCP server was configured to automatically assign IP addresses to other devices in the network. The server was configured manually via > desktop>IP configuration> static, entering an IP address -172.32.5. 50 , Subnet mask -255.255.0.0 and Default gateway -172.32.1.1. This ensures the server always has a consistent address in the network

After that DHCP service was enabled on the server by navigating to the DHCP configuration tab. A DHCP pool was created by specifying the IP address, Subnet mask, default gateway, starting IP address range for clients.



Step 11 -LAN 3

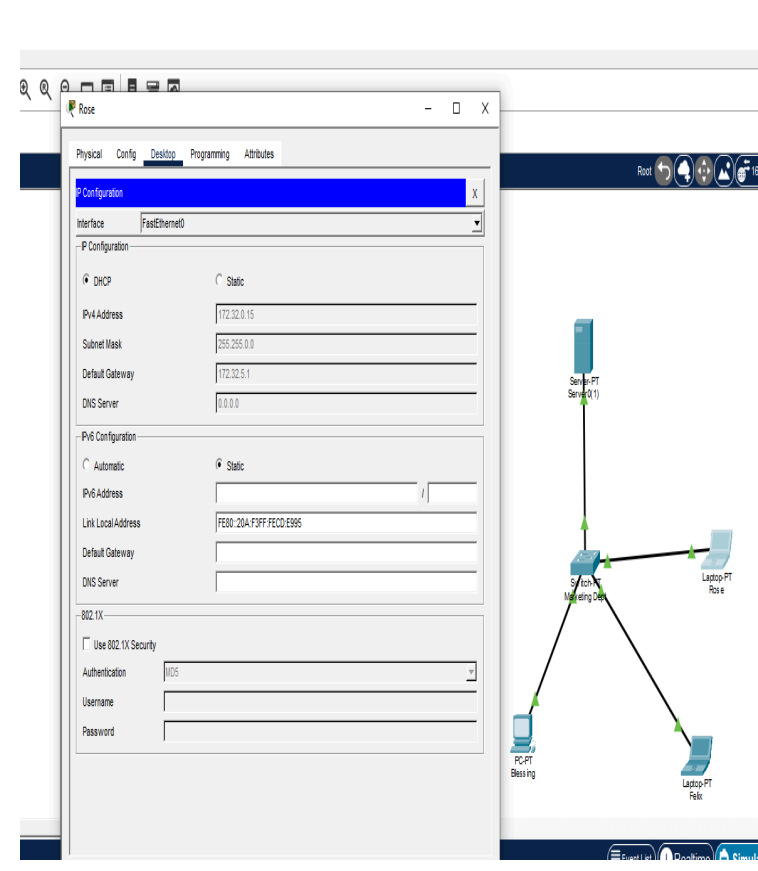
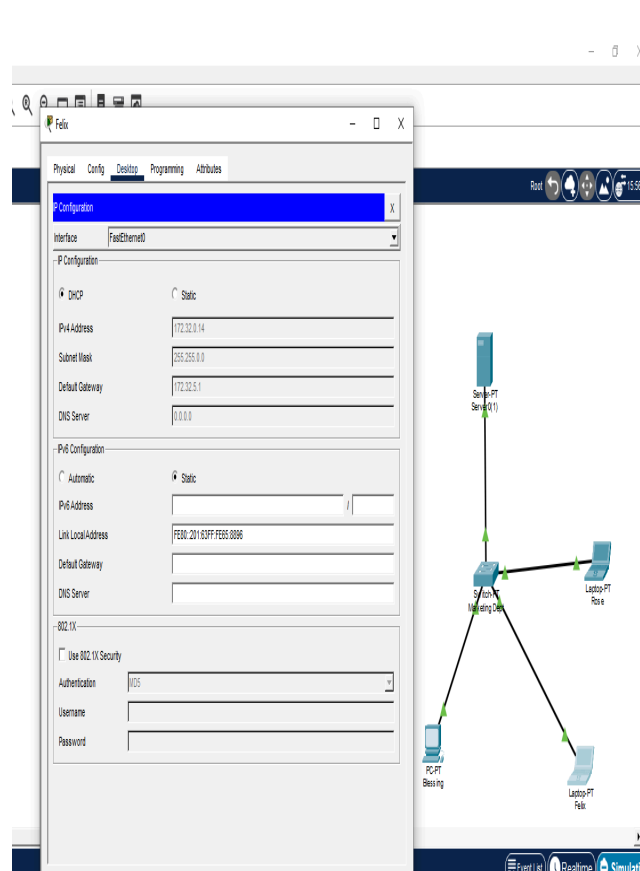
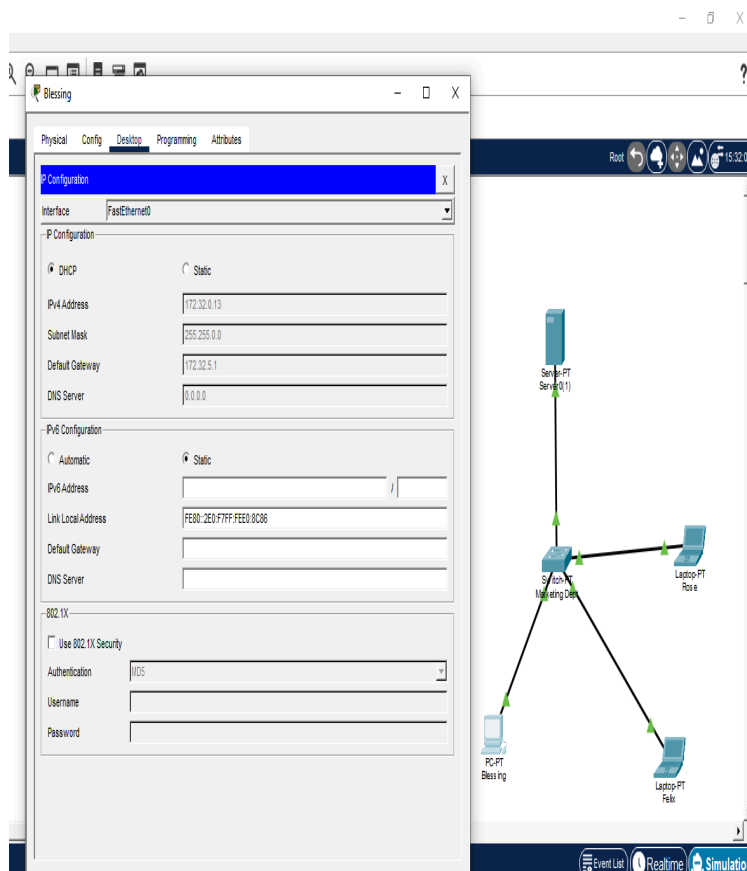
Verify DHCP IP Assignment:

All devices in LAN 3 were automatically assigned IP address, subnet mask, default gateway after the DHCP Configuration. Class B IP Address.

The first device(PC)
Successfully obtain IP address
Of 172.32.0.13 indicating proper
Communication with the DHCP
Server

The second device(laptop)
successfully obtain IP address
of 172.32.0.14 indicating proper
communication with the DHCP
server

The third device (Laptop)
obtained IP address of
172.32.0.15 indicating proper
communication with the
DHCP server

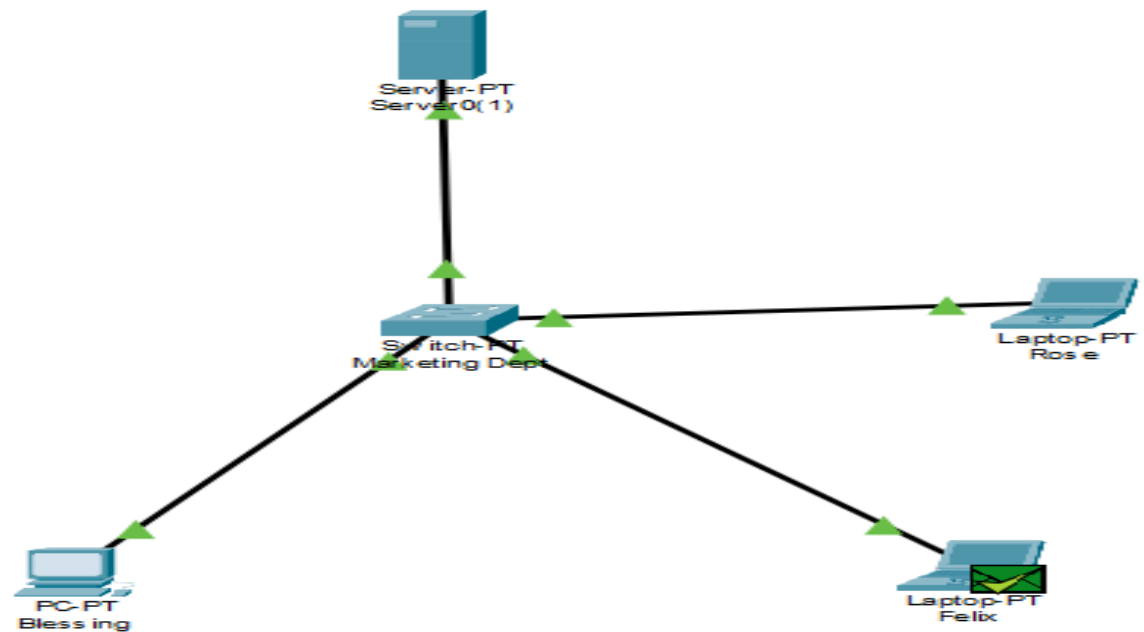


Step 9 LAN 3

Connectivity Test within LAN 3

To verify network communication within LAN 3, a connectivity test was performed by sending ping requests between devices.

Each device successfully responded to the ping, confirming that all hosts within the LAN 3 can communicate with each other.



Event List										
Realtime										
Simulator										
Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit	Delete	
Successful	Rose	Felix	ICMP		0.000	N	0	(e...	(delete)	
Successful	Felix	Blessing	ICMP		1.125	N	1	(e...	(delete)	

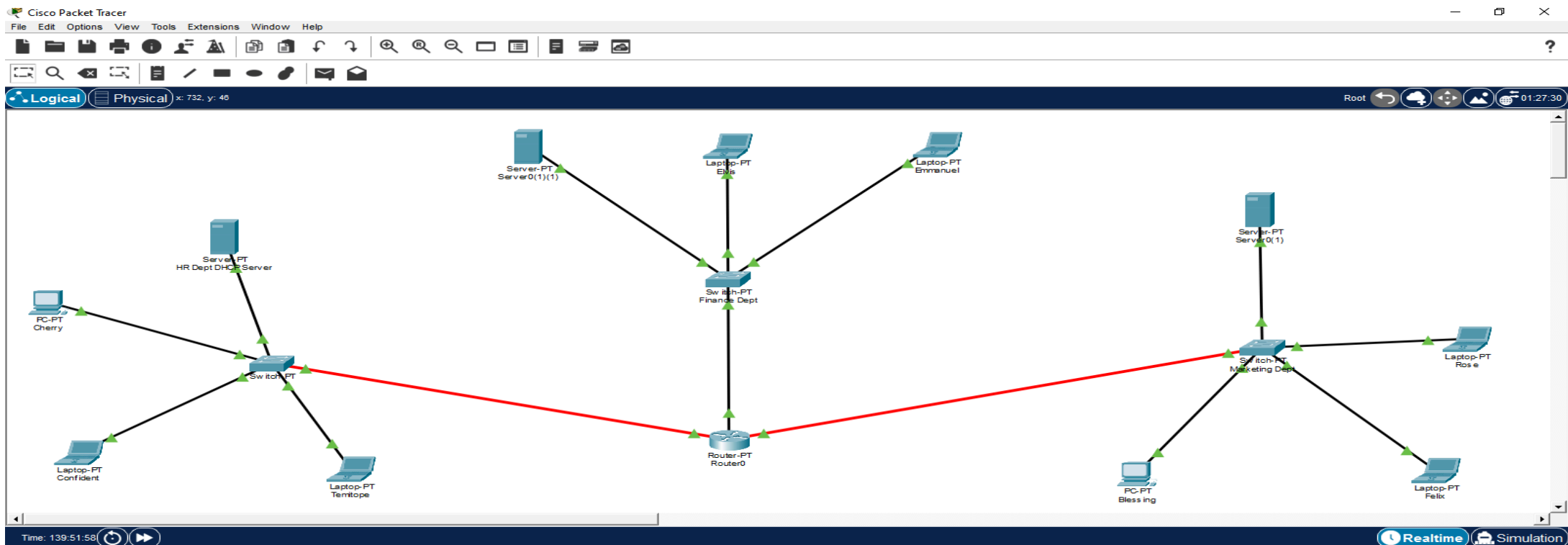
Step 12

Add Router to interconnect the Three-LANs:

After confirming successful communication within each individual's LAN through ping tests, a router was added to the network topology to enable communication between the three separate LANs. The router was selected from the Network Devices > Routers Category in Cisco Packet Tracer and placed centrally in the workplace to serve as the main routing device.

Each LAN switch was then connected to the router using appropriate copper straight-through cables through the router's available Ethernet interfaces. The router acts as the gateway for all the three networks, receiving traffic from on LAN and forwarding it to the correct destination LAN.

The router interface was later assigned IP addresses corresponding to each LAN subnet, allowing devices from different networks to communicate successfully. This step transforms the three independent LANs into one interconnected network while maintaining separate subnets.



Step 13

Configure the Router Using Settings

Each of the connected interfaces was selected and assigned an IP address and subnet mask based on the subnet of the LAN it was connected to.

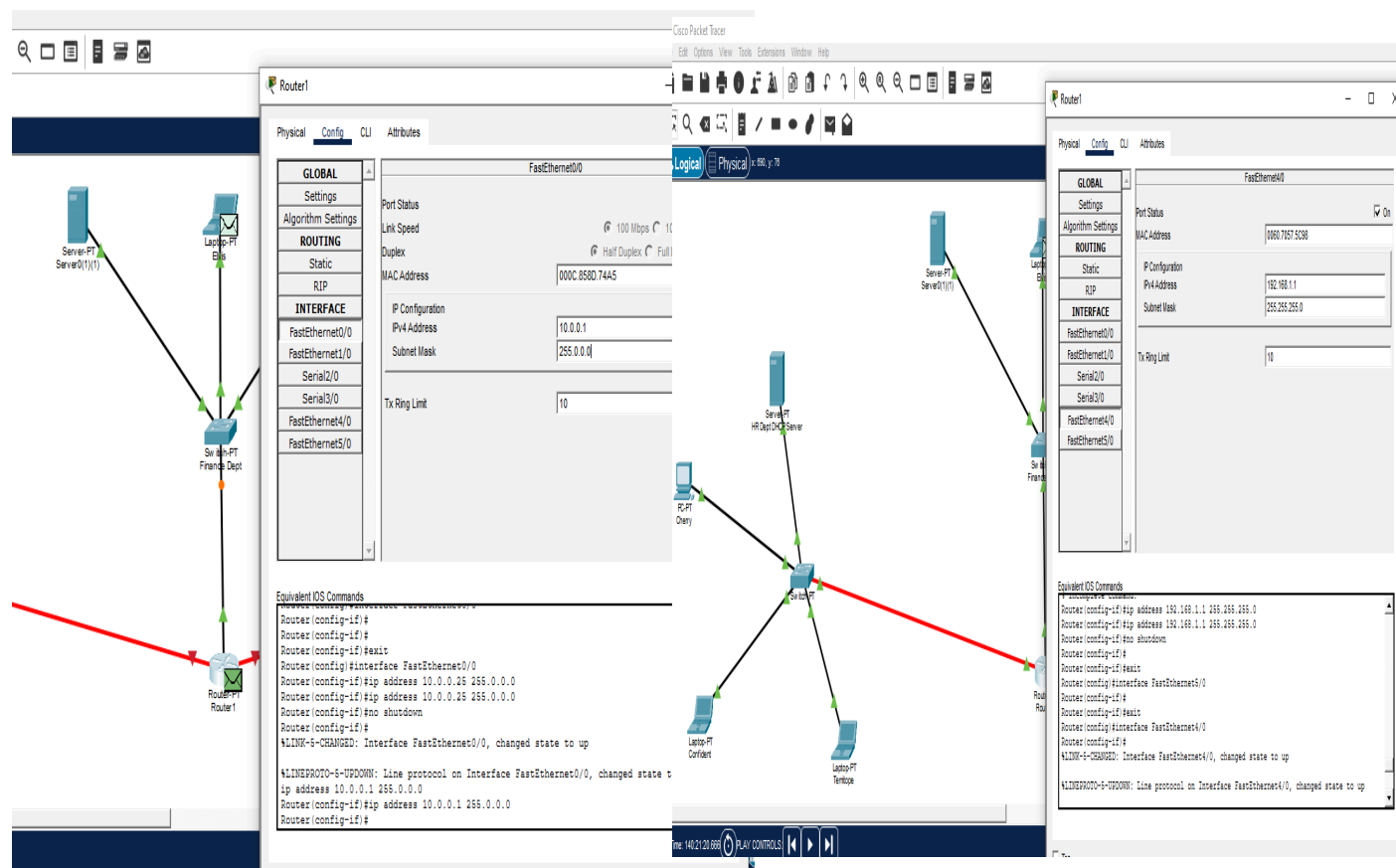
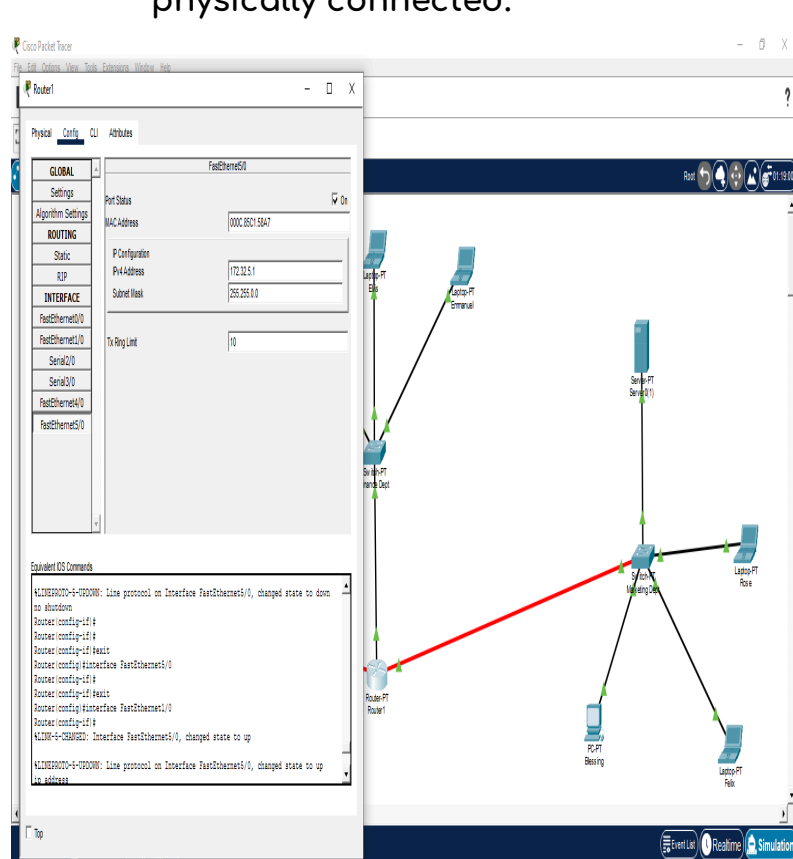
The router interface were configured as follows:

LAN 1 interface > FastEthernet4/0 > 192.168.1.1 with subnet mask 255.255.255.0

LAN 2 interface > FastEthernet0/0 > 10.0.0.1 with subnet mask 255.0.0.0

LAN 3 interface > FastEthernet5/0 > 172.32.1.1 with subnet mask 255.255.0.0

Each port was then switched On, in the settings panel to activate the interfaces. To verify successful activation, the CLI was checked, where the interfaces displayed State Up, confirming that the ports were operational and physically connected.

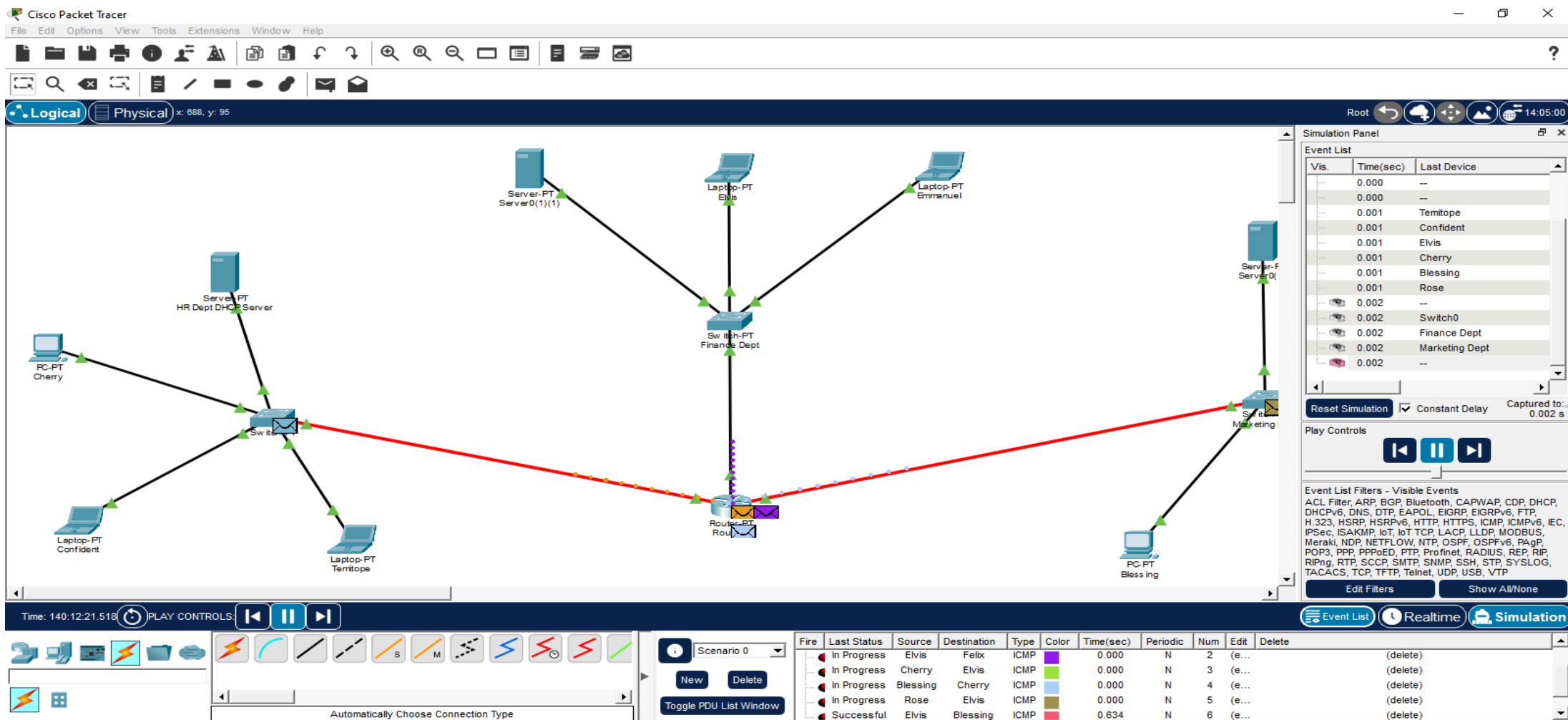


Step 14

Verify Inter-LAN Connectivity:

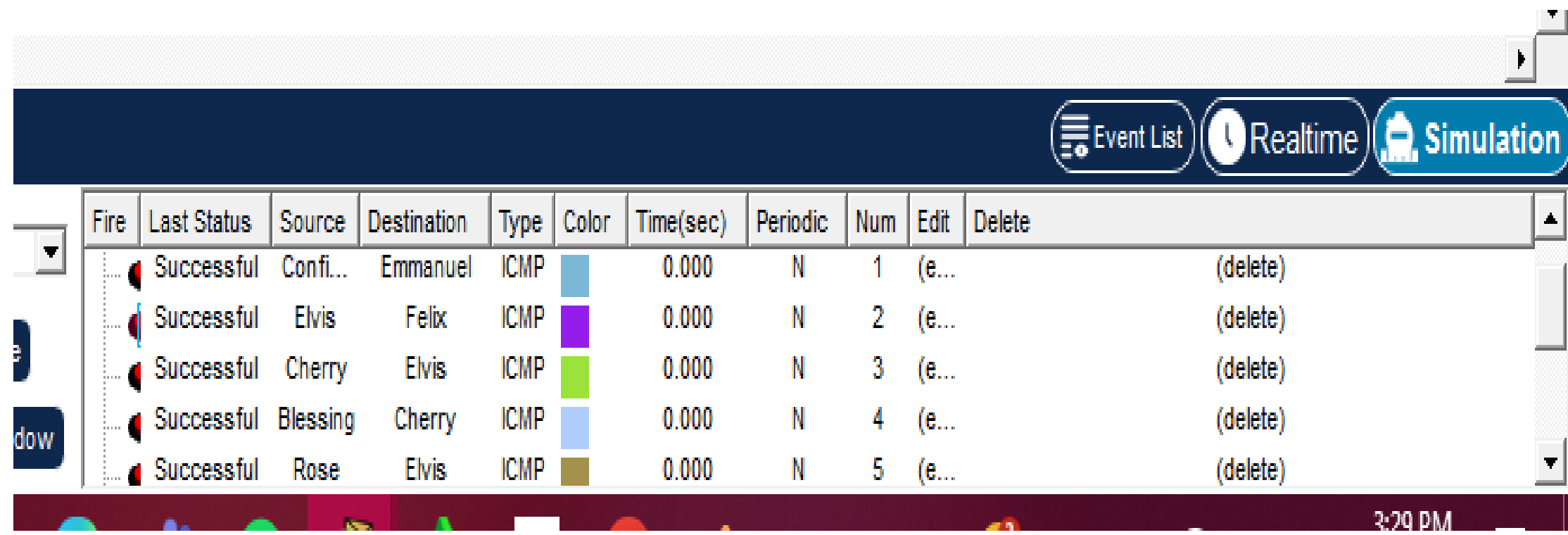
After configuring the router and assigning the correct gateway addresses, a connectivity test was carried out to confirm communication between devices located in different LANs. ping requests were sent from devices in one LAN network using the Add Simple PDU tool in the simulation mode.

The test results showed successful packet delivery between the networks, confirming that the router was correctly forwarding traffic from one subnet to another. This indicates that routing between LAN 1, LAN 2, LAN 3, was successfully established.



Conclusion

This project successfully demonstrated the design and implementation of a network consisting of three separate Local Area Networks (LANs) using Cisco Packet Tracer. The project achieved its objective of creating a functional multi-LAN network with dynamic addressing and successful routing. It also provided practical knowledge of network design, IP addressing, DHCP configuration, Router setup, troubleshooting and communication testing.



The screenshot displays the 'Event List' window in Cisco Packet Tracer. The window has a dark blue header with three buttons: 'Event List' (selected), 'Realtime', and 'Simulation'. Below the header is a table with the following columns: Fire, Last Status, Source, Destination, Type, Color, Time(sec), Periodic, Num, Edit, and Delete. The table contains five rows of event data, all with a status of 'Successful' and a time of '0.000'.

Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit	Delete
	Successful	Confi...	Emmanuel	ICMP		0.000	N	1	(e...	(delete)
	Successful	Elvis	Felix	ICMP		0.000	N	2	(e...	(delete)
	Successful	Cherry	Elvis	ICMP		0.000	N	3	(e...	(delete)
	Successful	Blessing	Cherry	ICMP		0.000	N	4	(e...	(delete)
	Successful	Rose	Elvis	ICMP		0.000	N	5	(e...	(delete)

At the bottom right of the window, the time '3:29 PM' is displayed.